

Ayanna Ewan

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Comp Sci 132

Dr. Krishna Kambhampaty

Final Report

This project is a two-player, terminal-based Snake & Ladder game. The two players enter their names and start at position 1 on a virtual board. They then take turns rolling a six-sided dice to move towards position 100. Along the way, the players might encounter snakes or ladders. If a player lands on a snake's head, they head down to its tail. If a player lands on the bottom of a ladder, they move to the top of the ladder. The first player to reach exactly position 100 wins the game. The program runs entirely in the terminal, interacts with users through text prompts, and enforces the rule that a player must land exactly on 100 to win.

The program is structured around a main game loop that alternates turns between two players. The game starts by asking for two names and placing the positions at 1, which are stored in a dictionary keyed by the player's name. A variable tracks whose turn it is, and after each turn, the other player takes control.

On each turn, the current player is told to press Enter in order to roll the dice. The dice roll is simulated using `random.randint(1, 6)`. The new position is calculated by adding the dice value to the player's current position. If this new position goes over 100, the move is rejected, and the player remains in place, enforcing the "exactly 100" rule. Otherwise, the player moves to the new position, and the program checks whether this position has a snake or a ladder. If it does, the player's position is updated accordingly, and a message is printed describing the event. After updating the position, the program checks if the player has reached 100. If yes, that player is declared the winner and the game ends.

The game is wrapped in a `main()` function that allows players to choose whether to play again. After a game finishes, the program asks if the users want another round. Input is validated so that only "y/yes" or "n/no" are accepted. If the players choose not to continue, the program ends.

The key data structures used in this project are dictionaries:

- Snakes = {16: 6, 48: 30}
 - The keys represent the positions of snake heads, and the values represent the positions of their tails. When a player lands on a key, their position is updated to the corresponding value.
- Ladders = {3: 22, 20: 38}
 - The keys represent the bottom of ladders, and the values represent the top positions. Landing on a key causes the player to climb to the value.

- A dictionary maps each player's name to their current position on the board. This makes it easy to manage multiple players and print their positions in a readable way.

One challenge was handling the rule that players have to reach 100 on the dot to win. A wrong code could allow players to go farther than 100. To fix this, the program looks at a player's position and checks if it exceeds 100. If it does, then the move is canceled, and the player stays at their previous position, with a message saying why.

Some improvements I could make to my game are to add more snakes and ladders. By expanding the dictionaries with more snakes and ladders would make the game more fun. I could add an option for there to be a custom number of players to make the game more inclusive. Another thing I could add is statistics at the end to show how many rolls it took for them to reach 100 and how many snakes or ladders they encountered.